

Stock Market Return Distribution Analyzer

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Project guide

Abstract—Historical stock prices show the direction of a market but not the shape or the risk of its returns. This paper presents the Stock Market Return Distribution Analyzer, an interactive analytical system that converts historical price data into daily, weekly and monthly returns and studies their distribution using the mean, median, variance, standard deviation, skewness and kurtosis. Histograms and distribution curves expose extreme values and unusual market movements, and a risk versus return view supports comparison. Stock analysis remains the core of the work. The analyzer is extended to cryptocurrency (Bitcoin and Ethereum) and to a market index (NIFTY 50) so that return behavior can be compared across financial asset classes and time horizons. The central research question is how return behavior and distribution differ across asset classes and time horizons. The methodology uses only standard, easily explained statistics and open Python tools, and the statistical modules are verified on synthetic data with known properties. The paper reviews ten recent studies on fat tails, volatility and multi asset return distributions, identifies the gap between research findings and simple reusable analysis tools, and describes the design, the evaluation plan and the expected outputs of the analyzer.

Keywords—return distribution, skewness, kurtosis, volatility, extreme movements, cryptocurrency, NIFTY 50, multi-horizon analysis, data science

I. INTRODUCTION

Stock prices are usually studied as charts, but investors and analysts make decisions about risk, and risk lives in returns rather than in prices. A price chart shows direction over time; it does not show how returns are distributed, how often large losses occur, or how one asset compares with another. Classical models often assume that returns follow a Normal distribution, yet empirical studies repeatedly find that large moves occur more often than a Normal distribution predicts [1], [2]. Describing return behavior therefore needs more than an average and a standard deviation: the asymmetry (skewness) and the tail weight (kurtosis) of the distribution must also be measured.

Three current trends make this description more urgent. First, retail participation has grown quickly. In India, the National Stock Exchange (NSE) reported more than 131 million unique registered investors by May 2026, and the total demat base reached about 231.5 million accounts by June 2026 [3]; many of these investors rely on mobile applications and simple charts rather than statistical tools. Second, digital assets have moved into regulated mainstream products. Spot Bitcoin exchange-traded funds (ETFs) were launched in the United States in January 2024 and spot Ether ETFs followed in July 2024 [4], and Bitcoin and Ether spot ETFs together drew about 31 billion US dollars of net inflows in 2025 [5]. Investors therefore compare stocks, market indices and cryptocurrencies side by side, although these assets differ in trading calendar and in volatility. Third, recent studies show that conclusions depend on the time horizon. Dehouche [6]

reported that lower sampling frequencies sharply reduced the kurtosis of Bitcoin log-returns, while the tails of gold and S&P 500 returns behaved in the opposite way. Liu and Tsyvinski [7] likewise reported that the skewness and kurtosis of Bitcoin returns change markedly across daily, weekly and monthly frequencies.

Much recent data science work on markets concentrates on forecasting with complex machine-learning models. Descriptive analysis of return distributions, however, remains the foundation of risk assessment, and it can be explained, checked and reproduced by a student or an investor. This paper therefore follows a deliberately simple and transparent route: standard statistics, standard plots and open-source tools.

This paper presents the Stock Market Return Distribution Analyzer, an interactive analytical system that converts historical prices into daily, weekly and monthly returns, measures their distribution, visualizes extreme movements and compares assets. Stock analysis is the core; cryptocurrency (Bitcoin and Ethereum) and a market index (NIFTY 50) are added to compare behavior across asset classes. The research question is: *How does return behavior and distribution differ across financial asset classes and time horizons?* The contributions are: (i) an integrated framework that combines return-distribution statistics, multi-period analysis and cross-asset comparison in one interactive system; (ii) a simple, explainable definition of extreme movement and a calendar-aware comparison procedure for assets that trade on different days; and (iii) a verification of the statistical modules on synthetic data with known properties, together with an evaluation plan for market data. Section II reviews the literature, Section III states the problem and objectives, Section IV describes the methodology, Section V the system design, Section VI the verification, Section VII the expected analysis and limitations, and Section VIII concludes.

II. LITERATURE REVIEW

A. Fat Tails and Non-Normality

Massing and Ramos [2] fitted several distribution models to daily and hourly log-returns of equity indices from different countries and compared the fits using the Kolmogorov–Smirnov and Anderson–Darling statistics and the AIC and BIC criteria. A mixture of three Student’s *t* distributions fitted best, and its components described the extreme, moderate and small returns. Welch [1] used historical U.S. stock returns to derive a simple adjustment to Z-scores for left-tail probabilities and showed that large losses occur more often than the Normal distribution predicts, with returns behaving roughly like a Student’s *t* distribution with two to five degrees of freedom. Eom et al. [8] measured tail fatness and found that diversification does not remove it: in Korean data the positive tail thins while the negative tail grows fatter as stocks are added, which helps to explain losses in crashes. Eom, Eom and Park [9] showed that the tail properties of individual Korean

stocks differ across the cross-section and must be considered when studying left-tail momentum.

B. Multiple Asset Types and Horizons

Wątopek et al. [10] analyzed 2017–2020 return distributions of currencies, cryptocurrencies and contracts for difference on stocks, indices and commodities, modelling the tails with power-law, stretched exponential and q-Gaussian functions. They found that the inverse-cubic power law is still a fair reference at short horizons and that the earlier trend towards faster convergence to normality is no longer seen. Sharma [11] built empirical return distributions of the Nifty 50 for horizons from one day to twelve years; one-day returns were mildly positively skewed with 56 percent of days positive, while one-year returns showed a 74 percent chance of gain. Studies of Bitcoin add that behavior depends on the sampling frequency [6], [7].

C. Volatility and Dependence in Emerging Markets

Mamilla et al. [12] applied ARCH and GARCH models to nine NSE indices before, during and after COVID-19. The Nifty Realty index was the most volatile, and the Bank, Metal and IT indices gave higher returns during COVID-19 than before. Mata et al. [13] fitted a multivariate Normal Inverse Gaussian distribution to BRIC stock market returns, found evidence of integration among these markets, and reported that Value at Risk (VaR) performed better with this distribution than with multivariate GARCH models.

D. Tail Measures and Portfolios

Liu et al. [14] introduced a quantile-based robust kurtosis measure and tested it on monthly returns of 39 country indices from 1996 to 2019. Higher robust kurtosis predicted lower future returns, mainly in developed markets, and robust skewness carried a positive premium, mainly in emerging markets. Kan and Lassance [15] showed that fat tails worsen the out-of-sample performance of sample mean–variance and global minimum-variance portfolios and that accounting for fat tails improves it.

E. Summary and Research Gap

Table I summarizes the reviewed studies. Three observations follow. First, the literature is rich in tail models and in forecasting or pricing methods, but each study typically centers on one market, one asset type or one modelling method. Second, the mean and standard deviation alone miss the asymmetry and fat tails that these studies emphasize [1], [8]. Third, the findings are published as research methods, not as simple reusable analyzers that a student or an investor can run on chosen assets and horizons. The gap addressed here is therefore not a new statistical model but the integration of return calculation, distribution statistics, visualization of extremes, multi-period analysis and cross-asset comparison into one interactive and explainable system.

TABLE I. REVIEWED STUDIES AND THEIR LIMITATIONS FOR THIS WORK

Study	Focus	Limitation
Massing and Ramos [2]	Fits distribution models to daily and hourly index log-returns	Model fitting; no interactive tool
Wątopek et al. [10]	Currencies, cryptocurrencies and contracts for difference, 2017–2020; tail models	Mixes asset types; no daily, weekly, monthly stock comparison
Mata et al. [13]	Multivariate Normal Inverse Gaussian model for BRIC returns; Value at Risk	Cross-market dependence; no single-stock visuals
Eom et al. [8]	Tail fatness under diversification (Korea)	Northeast Asian data; portfolio tail risk

Study	Focus	Limitation
Eom, Eom and Park [9]	Tail properties and left-tail momentum (Korea)	Single market; momentum focus
Mamilla et al. [12]	GARCH on nine NSE indices around COVID-19	Volatility forecasting; no visual distribution comparison
Welch [1]	Z-score adjustment for left-tail probabilities	Left tail only; no cross-stock comparison
Liu et al. [14]	Robust kurtosis, 39 country indices (working paper)	Country indices; a research method, not a tool
Kan and Lassance [15]	Fat tails and portfolio performance	Portfolio weights, not single-asset distributions
Sharma [11]	Nifty 50 distributions, one day to 12 years (preprint)	Single index; no stock-level comparison

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

The project addresses four connected problems:

- *Raw prices hide return behavior.* Historical prices alone do not show how returns are distributed or how much they fluctuate over time.
- *Risk needs more than an average.* Variance, standard deviation, skewness and kurtosis describe volatility, asymmetry and fat tails, but they are slow and error-prone to compute by hand.
- *Extreme movements are easy to miss.* Without histograms and distribution curves, extreme values and unusual market movements go unnoticed.
- *Assets and horizons are hard to compare.* Stocks, cryptocurrencies and market indices may behave differently, and behavior changes with the time horizon. Without a consistent method, comparison remains subjective.

A simple analysis system is therefore needed that turns price history into daily, weekly or monthly returns and organizes them for statistical analysis and comparison.

B. Research Question and Objectives

The research question is: *How does return behavior and distribution differ across financial asset classes and time horizons?* It is divided into three sub-questions. RQ1: How do the mean, volatility, skewness and kurtosis differ between stocks, cryptocurrency and a market index? RQ2: How do these measures change from daily to weekly to monthly returns, and do the returns move closer to a Normal shape? RQ3: How often, and how large, are extreme movements in each asset class? The objectives are to:

- compute daily, weekly and monthly returns from historical price data of stocks, cryptocurrency and NIFTY 50;
- organize the return data so that it is ready for statistical analysis;
- measure the mean, median, variance, standard deviation, skewness and kurtosis of the returns;
- visualize distributions with histograms and distribution curves to identify patterns, extreme values and unusual movements;
- compare return distributions and volatility across asset classes and across time horizons; and
- relate average return to volatility to explain how risk and return are connected.

IV. PROPOSED METHODOLOGY

A. Data and Asset Selection

Daily closing prices are collected from Yahoo Finance [16]. Stocks form the core of the study, and Table II lists the asset classes. The stocks are three to four NSE-listed large-cap companies from different sectors. Bitcoin and Ethereum represent cryptocurrency, and NIFTY 50 represents a market index. A common window of about eight years (for example, January 2018 to the latest complete month) is used so that every asset, including Ethereum, has data for the whole period; availability of each series will be confirmed before the final run. Eight years give about 2000 daily, 400 weekly and 96 monthly observations.

TABLE II. ASSET CLASSES USED IN THE STUDY

Asset class	Examples	Trading days	Quote unit
Stocks (core)	Three to four NSE large-cap stocks from different sectors	Weekdays	INR
Cryptocurrency	Bitcoin, Ethereum	Every day	USD
Market index	NIFTY 50	Weekdays	Index points

B. Preprocessing and Calendar Alignment

Duplicate dates are removed, missing prices are dropped rather than filled, and all series are trimmed to the common window. Cryptocurrency trades every day whereas stocks and the index trade only on weekdays, so daily returns of the two groups are not measured on identical calendars [6]. Daily statistics are therefore computed on each asset's own calendar and interpreted with this caveat, while weekly and monthly returns use the last available closing price of each week or month, which makes them directly comparable across assets. All analysis uses percentage returns, so the quote unit does not affect the results; the exchange-rate effect of comparing USD-quoted and INR-quoted assets is listed as a limitation.

C. Return Computation

The simple percentage return of a period is

$$R_t = (P_t - P_{t-1}) / P_{t-1} \quad (1)$$

where P_t is the closing price at the end of period t . For weekly and monthly returns, P_t is the last closing price of the week or month, so returns are computed from prices and not by adding daily returns. Simple returns are chosen because they are easy to interpret; for small returns they are close to log-returns, which many studies use [2], [6].

D. Descriptive Statistics

For n returns of an asset, the mean, standard deviation and standardized return are

$$\mu = (R_1 + R_2 + \dots + R_n) / n \quad (2)$$

$$s = \sqrt{[\sum (R_t - \mu)^2 / (n - 1)]} \quad (3)$$

$$z_t = (R_t - \mu) / s \quad (4)$$

and the skewness and excess kurtosis are

$$g = (1 / n) \sum z_t^3 \quad (5)$$

$$K = (1 / n) \sum z_t^4 - 3 \quad (6)$$

where the sums run over all n returns. Table III explains how each statistic is read. The software applies the usual small-sample corrections to (5) and (6) [17], [18]. Excess kurtosis is zero for a Normal distribution, so a positive value indicates heavier tails. The standard deviation s is used as the measure of volatility.

TABLE III. STATISTICS AND HOW THEY ARE READ

Statistic	Reading
Mean	Average return per period
Median	Middle return; less affected by extreme values
Std. deviation	Typical spread of returns; the volatility measure
Skewness	Asymmetry; negative means the larger moves are losses
Excess kurtosis	Tail weight relative to a Normal curve (zero)

E. Extreme Movement Analysis

A return is called an extreme movement when its standardized value satisfies $|z_t| > 3$, that is, when it lies more than three standard deviations from the mean. For a Normal distribution about 0.3 percent of returns fall outside this band, so the observed share of extreme movements can be compared directly with this benchmark. The analyzer also lists the five best and five worst returns of each asset and horizon, and overlays a Normal curve with the same mean and standard deviation on every histogram so that fat tails are visible.

F. Risk versus Return Comparison

Each asset and horizon is placed on a plot of mean return against standard deviation. A simple return-to-risk ratio

$$RR = \mu / s \quad (7)$$

summarizes the return earned per unit of volatility. It is not annualized and uses no risk-free rate, so it is used only to compare assets within the same horizon.

V. SYSTEM DESIGN AND IMPLEMENTATION

Fig. 1 shows the five stages of the analyzer. The pipeline is implemented in Python. Data handling uses pandas [17], the statistics use SciPy [18], and plots are drawn with Matplotlib or Plotly. A Streamlit dashboard lets the user choose the asset or assets, the return horizon (daily, weekly or monthly) and the time period, and then shows the statistics table, the histogram with the Normal curve, the extreme-movement summary and the risk versus return plot. Every output is produced by a standard formula given in Section IV, so each number on the dashboard can be traced and explained.

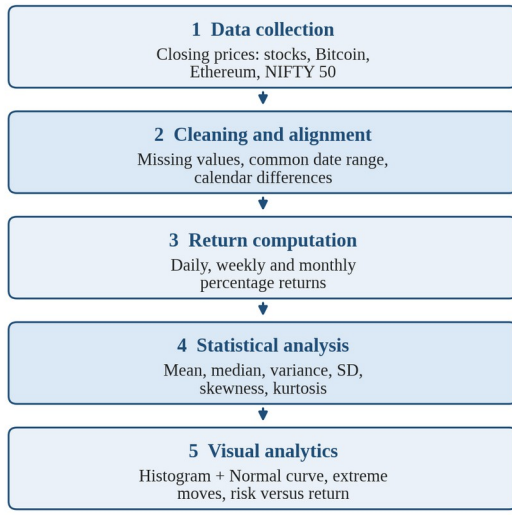


Fig. 1. Processing pipeline of the Stock Market Return Distribution Analyzer.

Optional extensions are kept separate from the core system and are implemented only if time permits: a Q–Q plot for visual checking of normality, the Jarque–Bera normality test, which uses skewness and kurtosis, and historical VaR, defined as the 5th-percentile return. Student’s *t* fitting, conditional VaR and comparison of sub-periods such as the COVID-19 period [12] are left as future work. Forecasting models, sentiment analysis and portfolio optimization are outside the scope of this work.

VI. VERIFICATION ON SYNTHETIC DATA

Before market data are analyzed, the statistical modules are verified on synthetic returns whose properties are known. Two series of 2016 daily returns (about eight years of 252 trading days) were generated with a fixed random seed and a daily standard deviation of 1 percent: one from a Normal distribution and one from a fat-tailed Student’s *t* distribution with five degrees of freedom, rescaled to the same standard deviation. Weekly and monthly series were formed by summing 5 and 21 consecutive daily values. These data are artificial and are used only to test the implementation; they are not market results.

TABLE IV. SYNTHETIC VERIFICATION OF THE STATISTICS

Data	Horizon	n	SD (%)	Skew	Ex. kurt.	>3 SD (%)
Normal	Daily	2016	1.00	0.00	0.09	0.35
Normal	Weekly	403	2.23	0.19	0.13	0.25
Normal	Monthly	96	4.53	−0.09	−0.03	0.00
Fat-tailed	Daily	2016	0.99	−0.07	2.66	1.04
Fat-tailed	Weekly	403	2.13	−0.14	0.27	0.25
Fat-tailed	Monthly	96	4.37	−0.15	−0.55	0.00

^a SD: standard deviation; Ex. kurt.: excess kurtosis; >3 SD: share of returns beyond three standard deviations.

Table IV shows that the modules behave as theory predicts. For the Normal series, skewness and excess kurtosis are close to zero and the daily share of extreme movements is 0.35 percent, near the 0.27 percent expected. For the fat-tailed series, the daily excess kurtosis is 2.66 and 1.04 percent of returns are extreme, about four times the Normal benchmark. Fig. 2 shows the corresponding analyzer output. When independent returns are summed over longer periods, excess kurtosis falls roughly in proportion to one divided by the number of periods summed, so the fat-tailed series becomes almost Normal at the weekly and monthly horizons (0.27 and −0.55). This gives a useful reference for the market study:

excess kurtosis that persists at weekly or monthly horizons in real data points to more than independent fat-tailed returns, for example volatility clustering or jumps.

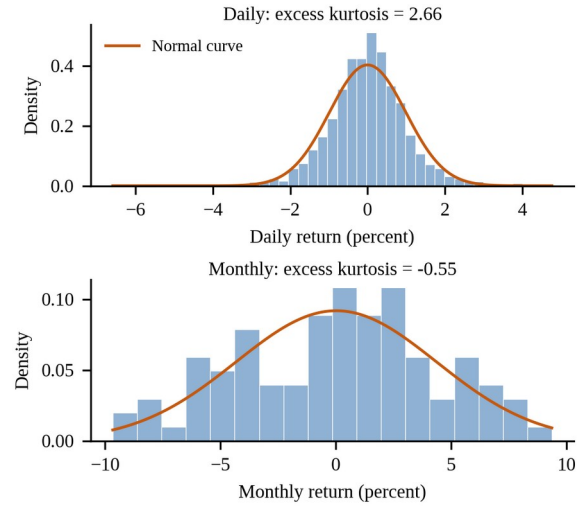


Fig. 2. Analyzer output on the synthetic fat-tailed series: daily and monthly histograms with a Normal curve.

The verification also shows how much sampling noise to expect. Over 2000 repeated Normal samples of 96 monthly observations, the middle 90 percent of the excess kurtosis values ranged from −0.65 to 0.89 and that of skewness from −0.41 to 0.41, although the true values are zero. For the fat-tailed daily series the 90 percent range of excess kurtosis was 1.73 to 11.33. Monthly statistics from about 100 observations, and kurtosis in general, must therefore be reported with this uncertainty in mind and not read as exact values.

VII. EXPECTED ANALYSIS AND DISCUSSION

A. Reading the Results

The analyzer will be run on the assets in Table II, and the results will be read against the three sub-questions. For RQ1, the statistics tables of the three asset classes are compared at each horizon. Cryptocurrency is expected to show larger dispersion than stocks and the index [6], [7], and the index is expected to be less volatile than a typical single stock because it averages many stocks; both are hypotheses to be tested and not assumptions. For RQ2, the change of skewness, kurtosis and standard deviation from daily to monthly returns is compared across asset classes. No single outcome is presumed, because the literature reports mixed evidence: excess kurtosis of Bitcoin fell with longer horizons while the opposite tail behavior was reported for S&P 500 [6], and convergence to normality is not always observed [10]. For RQ3, the share of returns beyond three standard deviations and the worst returns are compared with the 0.3 percent Normal benchmark and with the synthetic reference of Section VI.

B. Limitations

The study is descriptive and makes no forecast. Its findings apply to the chosen period and assets, and market regimes such as the COVID-19 period [12] may influence the statistics. Daily comparison between cryptocurrency and stocks is affected by the different trading calendars, and comparison of USD-quoted and INR-quoted assets ignores exchange-rate movements. Monthly samples are small (Section VI), and the standard deviation and kurtosis can be dominated by a few extreme observations when tails are heavy [6]. Data quality depends on the public data source. These limits will be stated with every reported result.

C. Practical Value

For students and beginner investors, the analyzer turns a price history into a small set of interpretable numbers and plots. For researchers, it offers a reproducible baseline analysis of return distributions across asset classes and horizons before any advanced model is applied.

VIII. CONCLUSION AND FUTURE WORK

This paper presented the Stock Market Return Distribution Analyzer, which integrates return-distribution statistics, multi-period analysis and cross-asset comparison in one interactive and explainable system. Stocks remain the core, and cryptocurrency and NIFTY 50 extend the analysis to answer the question of how return behavior and distribution differ across asset classes and time horizons. The design, the definitions and the calendar-aware comparison procedure were described, and the statistical modules were verified on synthetic data. The next step is the full run on market data, whose results will answer RQ1 to RQ3. Future work includes the optional Q–Q plot, Jarque–Bera test and historical VaR, Student’s *t* fitting, conditional VaR, sub-period comparison and a wider set of assets.

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